

Mesh charge-injection timing study

n_TOF X17 Micromegas γ -flash countermeasure — delay scan and retiming

X17 DAQ (automated study)

2026-07-11

Abstract

The prompt γ -flash saturates the Micromegas detectors at every proton pulse. A charge-injection pulse applied to the detector meshes (Module 6 section B of the N1081B trigger crate) is meant to counteract it, but was found firing ~ 760 ns *before* the flash, with no overlap. We scanned the injection delay using flash-triggered DREAM runs and found that placing the injection ~ 180 ns before the flash rise suppresses the event-averaged flash amplitude on detector C by $\sim 45\%$, while an injection arriving exactly at the rise gives no suppression. The delay knob (M6.B in0 Gate&Delay) was retimed from 500 ns to **1260 ns**.

1 Setup and method

Injection chain. PS/T0 $\rightarrow$ M6 (.245) section B fanout $\rightarrow$ four outputs (500 ns monostable) $\rightarrow$ Micromegas meshes. The timing knob is the **input Gate&Delay of M6.B channel 0** (gate=50 ns, delay=the scanned parameter; 5 ns steps). The injected pulse rails detector A's front-end, couples moderately to B/C, and is therefore directly visible in the DREAM waveforms.

Measurement. Flash-triggered DREAM runs (run mode 1: M4.D=OR(lemo0)=PS line; 400 samples $\times$ 20 ns, latency 60; ~ 79 events per 4 min at the parasitic pulse rate). For each delay setting the event-averaged, channel-averaged waveform per FEU (`check_flash.py` profiles) gives the flash amplitude and the injection position. FEU $\rightarrow$ detector map: A=03/04, B=05/06, C=07/08, D=01/02 (x,y).

Runs.

variant	M6.B in0 delay	time (2026-07-11)	pulses used	$\langle I \rangle$ [10^{10} p]
m1_lat60	500 ns (original)	12:35	89	675
m6_inj_aligned	1260 ns	15:48	130	627
m6_inj_aligned2	1440 ns	15:55	95	633

The two decisive runs (1260 vs. 1440 ns) had beam intensities within 1% of each other, so the amplitude difference below cannot be a beam effect.

2 Results

Detector-C flash amplitude (event-averaged peak – baseline, FEU 07) vs. injection delay:

M6.B in0 delay	injection vs. flash rise	det C peak–base [ADC]	change
500 ns (original)	~ 760 ns early, separate pulse	3099	—
1260 ns (chosen)	~ 180 ns before the rise	1709	–45%
1440 ns	at/just after the rise	3312	+7%

Figures 1–3 show the underlying waveforms.

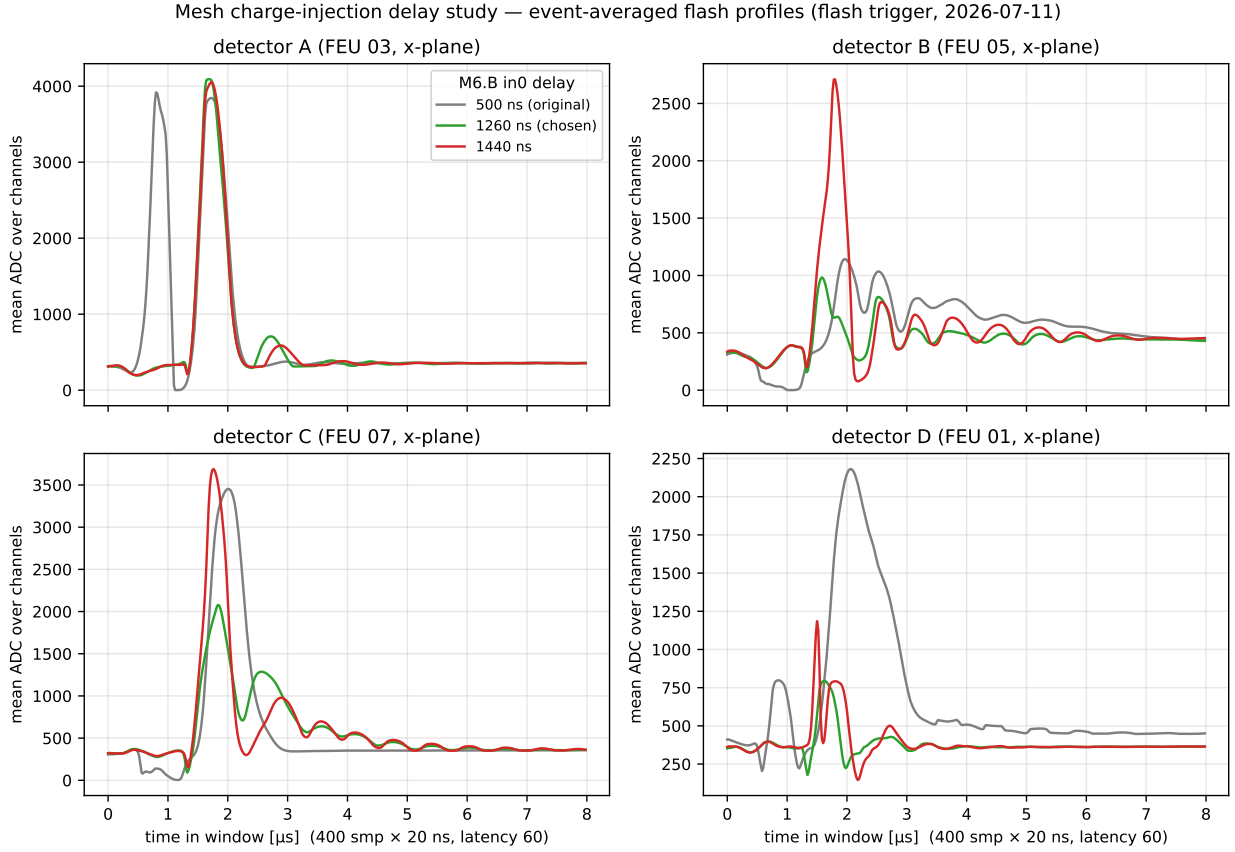


Figure 1: Event-averaged, channel-averaged waveforms per detector for the three delay settings. Gray: original 500 ns — the injection is the separate railed pulse at $\sim 0.8 \mu\text{s}$, the flash follows at $\sim 1.6\text{--}2 \mu\text{s}$. Green: 1260 ns — injection merged into the flash rise; detector C’s flash peak drops by 45%. Red: 1440 ns — injection at the rise; no suppression (det C), and detector B’s peak is *enhanced* (constructive overlap). Detector A rails at 4095 in all settings (strongest injection coupling). **Caveat:** the detector-D panel is not comparable across runs — its amplification HV was 640 V during the 500 ns run and 100 V (parked after a trip) during the other two.

3 Interpretation and recommendation

- The cancellation needs the mesh to be *pre-loaded*: an injection arriving ~ 180 ns *before* the flash rise suppresses the recorded flash by $\sim 45\%$ (det C), whereas an injection at the rise does nothing (det C) or even adds constructively (det B). “Right at the beginning of the flash” is therefore best implemented as *slightly before* it.
- The response is detector-dependent (coupling and polarity differ per mesh): A rails regardless; C benefits most; B prefers earlier injection as well (its 1440 ns peak nearly triples the 500 ns one).
- **Operating point set: M6.B in0 delay = 1260 ns** (as-left, snapshot `dump_2026-07-11_veto_injection_`
- The lever arm is exactly 1 ns per ns (input G&D). A finer scan (1100–1350 ns in 50 ns steps, one 4-min flash run each) would locate the suppression optimum per detector; amplitudes should be normalised to the beam intensity logged in `beam_monitor/logs/`.

Provenance

Runs and figures: `~/beam_july/test/latency_singles/` (`m1_lat60`, `m6_inj_aligned`, `m6_inj_aligned2`; figure script `make_mesh_report_figs.py`). Trigger/readout context: `n1081b/RUN_MODES_2026-07.md` and `n1081b/HANDOFF_2026-07-11_latency_tuning.md`. Board state: `n1081b/snapshots/dump_2026-07-11_vet`

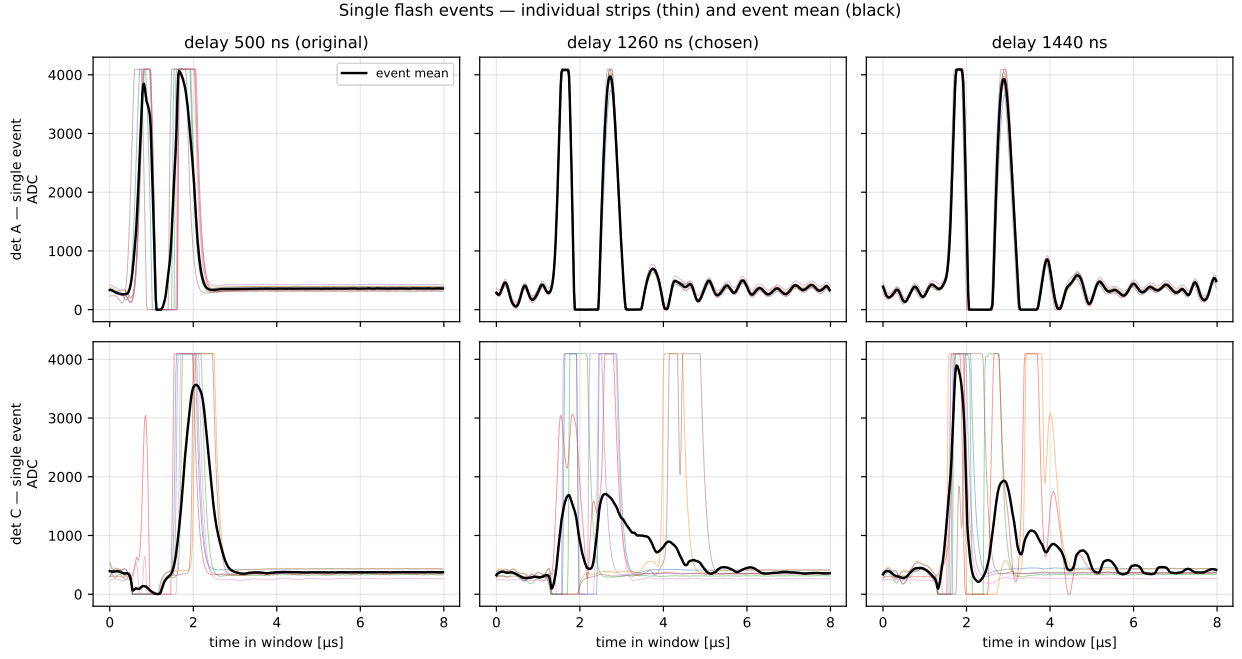


Figure 2: Single flash events (detectors A and C): individual strips (thin, subset) and the event mean (black). At 500 ns detector A shows the two separate railed pulses (injection, then flash). At 1260/1440 ns they merge. The hard clips to ADC 0 are the coherent post-pulse undershoot.

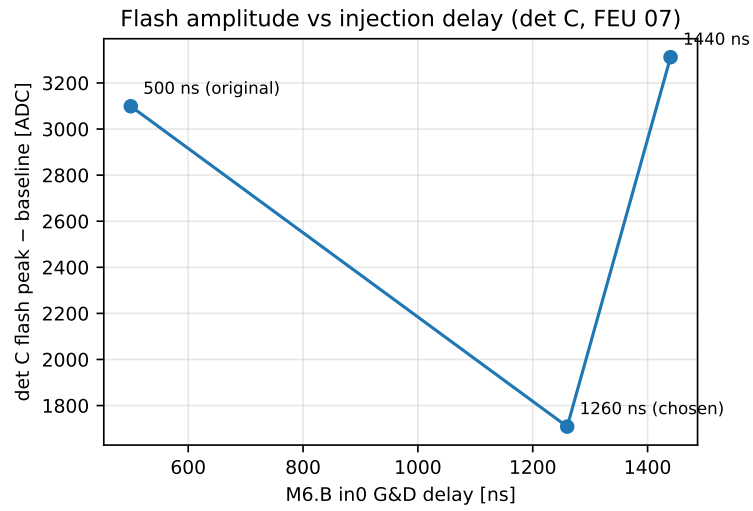


Figure 3: Detector-C event-averaged flash amplitude vs. injection delay.